SM PHARMACEUTICALS SDN. BHD.

IBUSPAN TABLET
(IBUPROFEN TABLET 200 MG)

Description:
Pink, round shaped, biconvex, film coated tablet.

Composition:
Each tablet contains: Ibuprofen 200 mg

Actions and Mode or mechanisms of action:
Ibuprofen has shown anti-inflammatory, antipyretic, and analgesic activity in both animals and humans. Higher doses are required for anti-inflammatory effects than for analgesia. The anti-inflammatory action of ibuprofen may be due to inhibition of synthesis and / or release of prostaglandins. Ibuprofen probably produces anti-pyresis by acting on the hypothalamus, with heat dissipation being increased as a result of vasodilation and increased peripheral blood flow. Animal studies indicate that ibuprofen is a peripherally acting, not a centrally acting, analgesic. The drug does not possess glucocorticoid or adrenocorticoid properties and has no uricosuric action.
In humans, ibuprofen causes gastric mucosal abnormalities including oedema, erythema, and submucosal petechial hemorrhages and erosion. In one study, Ibuprofen, in a dosage of 1.2 g daily, produced less severe gastric mucosal dosage of 100 mg daily. Ibuprofen inhibits platelet aggregation and prolongs bleeding time but does not affect prothrombin time or whole blood clotting time.

Pharmacology:
Ibuprofen is approximately 80% absorbed from the gastro-intestinal tract and peak plasma concentrations occur about 1 to 2 hours after ingestion. Ibuprofen is extensively bound to plasma proteins and has a half-life of about 2 hours. It is rapidly excreted in the urine mainly as metabolites and their conjugates. About 1% is excreted in urine as unchanged ibuprofen and about 14% as conjugated ibuprofen.

Indications:
Ibuprofen is used in the management of mild to moderate pain in conditions such as dysmenorrhoes, migraine, postoperative pain, ankylosing spondylitis, osteoarthritis, and rheumatoid including juvenile rheumatoid arthritis, and in other musculoskeletal and joint disorders such as sprains and strains

Contraindications:
Ibuprofen is contraindicated in patients with known hypersensitivity to the drug and in patients in whom bronchospasm, angioedema, or nasal polyps are precipitated by aspirin

or other NSAIDs. NSAIDs generally are contraindicated in patients in whom urticaria, angioedema, bronchospasm, severe rhinitis, or shock is precipitated by aspirin or other NSAIDs, although the drugs have occasionally been used in NSAID-sensitive patients who have undergone de-sensitization. Patients who are considering use of ibuprofen for self-medication should be advised not to take the drug if they have expierenced a severe allergic reaction (e.g., asthma, edema, shock, urticaria) to aspirin or other NSAIDs. Should not be administered to patients with a history of, or active peptic ulceration.

Side Effects / Adverse Reactions:

The most frequent adverse effects occurring with ibuprofen are gastro-intestinal disturbances; reactions range from abdominal discomfort, nausea and vomiting, and abdominal pain to serious gastro-intestinal bleeding or activation of peptic ulcer. CNS-related side effects include headache, dizziness, nervousness, tinnitus, depression, drowsiness, and insomnia. Hypersensitivity reactions may occur occasionally and include fever and rashes. Hepatotoxicity and aseptic meningitis which rarely may also be hypersensitivity reactions. Ibuprofen can provoke bronchospasm in patients with asthma. Ibuprofen and other non-steroidal anti-inflammatory drugs may cause cystitis and haematuria. They may also cause renal failure, interstitial nephritis, and nephrotic syndrome.

Other adverse effects include anaemias, thrombocytopenia, neutropenia, eosinophilia, agranulocytosis, abnormalities in liver function tests, blurred vision, changes in visual colour perception, and toxic amblyopia.

Hypersensitivity – assorted skin disorders including rashes of various types, pruritus, urticaria, purpura angioedema, photosensitivity and less commonly bullous dermatoses.

Precautions / Warnings:

Ibuprofen or other non-steroidal anti-inflammatory drugs should not be given to patients with active peptic ulceration. It should be given with care to the elderly, to patients with asthma or bronchospasm, bleeding disorders, cardiovascular disease, a history of peptic ulceration, and in liver or renal failure. Patients with congestive heart failure, cirrhosis, diuretic-induced volume depletion, or renal insufficiency require local synthesis of vasodilating prostaglandins to maintain renal perfusion, and therefore these patients are at greater risk of developing renal dysfunction due to NSAID-induced inhibition of renal prostaglandin synthesis.

Care is required in those who are also receiving coumarin anticoagulants.

Patients who are sensitive to aspirin or other NSAIDs should generally not be given ibuprofen.

Ibuprofen should be discontinued in patients who experienced blurred or diminished vision, or changes in colour vision. Patients with collagen disease may be at increased risk of developing aseptic meningitis.

Pregnancy – use of ibuprofen during the second half of pregnancy is not recommended because of possible adverse effects on the fetus, such as premature closure of the

ductus arteriosus, which may lead to persistent pulmonary hypertension in the newborn.
Use during pregnancy should, if possible, be avoided.
Breast-feeding – Problems in humans have not been documented with most of the NSAIDs.
Studies in humans have failed to detect ibuprofen in breast milk using methodology capable of detecting the medication in a concentration of 1 mcg /ml.

Drug Interactions:
Anticoagulants and Thrombolytic Agents – In several short-term, controlled studies. Ibuprofen did not have a substantial effect on the prothrombin time of patients receiving oral anticoagulants; however, because Ibuprofen may cause GI bleeding, inhibit platelet aggregation, and prolong bleeding time and because bleeding has occurred when Ibuprofen and coumarin-derivative anticoagulants were administered concomitantly, the drug should be used with caution and the patient carefully observed if the drug is used concomitantly with any anticoagulant (e.g. warfarin) or thrombolytic agent (e.g., streptokinase)

Other Nonsteroidal Anti-inflammatory Drugs– In animal studies, blood concentrations of ibuprofen increased when Aspirin and ibuprofen were administered concomitantly. Although limited studies have not shown decreased blood concentrations of ibuprofen when these drugs were administered concurrently, this possibility should be considered. In addition, concomitant administration of ibuprofen and salicylates, phenylbutazone, indomethacin or other NSAIDs could potentiate the adverse GI effects of these drugs, and thus ibuprofen probably should not be administered with these agents

Digitalis (Cardiac) glycosides – Ibuprofen has been shown to increase serum digoxin concentrations, increased monitoring and dosage adjustments of the digitalis glycoside may be necessary during and following concurrent NSAID therapy.

Quinolone antibiotics – Ibuprofen can increase the central adverse effects of quinolone antibiotics leading to convulsions.

Lithium – Ibuprofen has been reported to increase the steady-state concentration of lithium, possibly by decreasing its renal clearance; increased monitoring of lithium concentrations is recommended during and following concurrent use.

Methotrexate – Ibuprofen may decrease protein binding and / or renal elimination of methotrexate, resulting in increased and prolonged methotrexate plasma concentrations and an increased risk of toxicity, especially during high-dose methotrexate infusion therapy; fatalities have been reported; it is recommended that NSAID therapy be withheld for varying periods of time, depending on the elimination half-life of the individual NSAID [12 to 24 hours for agents with a short elimination half-life to up to 10

or 12 days for agents with a very long elimination half-life] prior to administration of a high-dose methotrexate infusion; also, NSAID therapy should not be resumed following the infusion until the methotrexate plasma concentration has decreased to a non-toxic level, usually at least 12 hours.
Severe, sometimes fatal, methotrexate toxicity has also been reported when NSAIDs were used concurrently with low to moderate doses of methotrexate, including doses commonly used in the treatment of rheumatoid arthritis or psoriasis; caution in concurrent use is recommended, with dosage of methotrexate being adjusted as determined by monitoring the plasma methotrexate concentration and / or adequacy of the patient's renal function)

Corticosteroids – Concurrent use with an NSAID may increase the risk of gastrointestinal side effects, including ulceration or hemorrhage; however, concurrent use with a glucocorticoid or corticotropin in the treatment of arthritis may provide additional therapeutic benefit and permit reduction of glucocorticoid or corticotropin dosage.

Recommended Dosage, Dosage Schedule and Route of Administration:
Route of Administration: For Oral use
The usual dose by mouth is 1.2 to 1.8 g daily in divided doses although maintenance doses of 0.6 to 1.2 g daily may be effective in some patients. If necessary the dose may be increased to 2.4 g daily. If gastro-intestinal disturbances occur, Ibuprofen should be given with food or milk.
A suggested dose for children is 20 mg per kg body-weight daily in divided doses with up to 40 mg per kg being given daily in juvenile rheumatoid arthritis if necessary; the maximum daily dose has been started to be 500 mg for those weighing less than 30 kg.
For children dose. Not recommended for children less than 7 kg body weight.

Symptoms and Treatment for Overdose, and Antidote(s):
Drowsiness was the only adverse effect experienced by a child 1 year of age who digested 1.2 g (120 mg / kg) of the drug even though the blood ibuprofen concentration 90 minutes after ingestion was 700 ug/ml (nearly 10 times the highest concentration previously recorded in adults following a single oral 800 mg dose of the drug). Signs of acute intoxication or late sequelae also were not present following ingestion of a 120 mg/kg dose of the drug in another child. However, in a 12 kg child who ingested 2.8 – 4 g of the drug, apnea, cyanosis, and response only to painful stimuli were present about 1.5 hours after ingestion; respiration could be induced by painful stimuli. The blood ibuprofen concentration in this child was about 103 ug/ml at about 8.5 hours after ingestion and the child appeared to have completely recovered at 12 hours following treatment with oxygen, sodium bicarbonate, and parenteral infusion of dextrose and 0.9% sodium chloride. Dizziness and nystagmus occurred in a 19 years old who ingested 8 g of the drug over a period of a few hours.
Treatment

When acute overdosage of ibuprofen occurs, the stomach should be emptied by inducing emesis or by lavage, particularly if there is evidence that the drug has been ingested recently (within 1 hour), and standard measures to maintain urine output should be instituted. Since ibuprofen is acidic and is excreted in the urine, forced alkaline diuresis, might be beneficial.

WARNINGS
RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID
Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy.
Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.
Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Storage Conditions, User Instructions and Pharmaceutical Precautions:
KEEP ALL MEDICINES OUT OF REACH OF CHILDREN.
Store below 30°C in closed airtight container.
Protect from direct light and heat.

Packing / Pack Sizes:
Blister-pack of 10’s, 10x10’s/box and 100x10’s/box.

Shelf life: 3 years

Manufacturer and Product Registration Holder:
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